

Perfect Vessel Viewer Upgrade Plan

Segment Project

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Status: technical upgrade plan for the coronary vessel visualisation proof of concept.

Perfect Vessel Viewer Upgrade Plan

Goal

Build a robust coronary vessel viewer that can start from either expert MEDIS contours or a voxel plaque/wall mask, then produce aligned 3D, axial, CPR cross-section, and straightened MPR views for presentation, review, and later quantitative work.

The central design idea is a unified vessel model. MEDIS provides sparse expert contour rings; vendor plaque masks provide dense voxel labels. Both should be converted into the same internal representation:

centerline + local frame + lumen surface + wall surface + optional plaque

labels.

Every view, export, and validation check should then operate on this unified model rather than on one source format directly.

Canonical VesselScene Contract

The most important architecture decision is to freeze a common scene contract before building a larger app. The backend or preprocessing pipeline produces a `VesselScene`; the frontend consumes it. MEDIS and voxel-mask inputs must both emit the same object shape.

```
VesselScene = {
  patient_id: string,
  vessel_name: string,
  source_type: "medis" | "mask" | "hybrid",
  centerline: {
    points: number[][],          // N x 3 world-mm points
    tangents: number[][],
    normals: number[][],
    binormals: number[][],
    arc_length: number[],
    station_count: number
  },
  surfaces: {
    lumen: Mesh,
    wall: Mesh,
    plaque?: Mesh
  },
  volumes: {
    cpr_cross: NiftiVolume,
    cpr_long: NiftiVolume
  },
  overlays: {
    medis_rings?: PolylineSet,
    mask_contours?: PolylineSet
  },
  quant: {
    per_station: PerStationMetrics,
    lesion_summary: LesionSummary
  },
  provenance: {
    ct_uid: string,
    spacing: number[],
    origin: number[],
    direction: number[],
    algorithm_versions: object,
```

```

    created_at: string
  }
}

```

This contract is the boundary between compute and presentation. Once it is stable, a MEDIS converter, a vendor-mask converter, and a web viewer can be developed independently.

Glossary

Term	Meaning in this project
CTA	Coronary CT angiography volume used as the intensity image source.
MEDIS	Expert annotation/export format containing lumen and vessel-wall contour rings.
Lumen / inner wall	The contrast-filled inside of the vessel; the boundary used for diameter and stenosis measurements.
Vessel wall / outer wall	The outer vessel boundary around the lumen and plaque/wall tissue.
Plaque mask	Voxel label map describing plaque or wall tissue, usually between lumen and outer vessel wall.
Centerline	A 3D curve running through the vessel centre from proximal to distal.
Station s	Integer position along the resampled centerline; all linked views should use the same station index.
CPR	Curved Planar Reformation: a CT reformat that follows the curved vessel path instead of a fixed axial/coronal/sagittal plane.
CPR cross-section	A slice perpendicular to the vessel centerline at station s ; used to inspect lumen/wall shape locally.
MPR	Multiplanar Reformation: an arbitrary 2D slice or slab sampled from a 3D CT volume.
Straightened MPR	A longitudinal CPR view where the curved vessel is unfolded into a straight strip from proximal to distal.
RMF	Rotation-Minimizing Frame: local tangent/normal/binormal axes transported along the centerline with minimal twist.
FC03 / FC51	CT reconstruction kernels in this dataset. FC03 is smoother and useful for axial context; FC51 is sharper and better for vessel detail.

Term	Meaning in this project
Window / level	Display mapping from HU values to grayscale; for the slide we use W1000/L350 (-150 . . 850 HU).
World coordinates	Physical millimetre coordinates from DICOM/NIfTI metadata, used to align CT, MEDIS, masks, and meshes.
Voxel coordinates	Integer or fractional array indices used when sampling the CT or mask volume.
Annulus mask	Binary/label mask containing the vessel-wall/plaque shell between inner lumen and outer wall.
FMM	Fast Marching Method used to find minimal-cost vessel paths through a distance/radius field.
glTF / GLB	Web-native 3D mesh format; .glb is the binary single-file form preferred for browser loading.
Scene bundle	A zip archive containing the VesselScene manifest, volumes, meshes, overlays, quant files, validation report, and preview image.
Taubin smoothing	Mesh smoothing without the volume shrinkage typical of plain Laplacian smoothing ¹ .

Current Presentation Prototype

- Input: MEDIS lumen and vessel-wall contour rings for patient 01-BER-0088, vessel LAD.
- CT: DICOM converted to NIfTI; FC03 for axial context, FC51 sharp-kernel reconstruction for CPR/MPR detail.
- Display convention: axial CTA is shown with anterior up after accounting for NIfTI direction/orientation.
- Outputs:
 - 3D lumen/wall mesh from contour rings.
 - Separate 3D point clouds for inner/lumen and outer/wall contours.
 - Straightened MPR with lumen and wall boundary envelopes.
 - CPR cross-sections with MEDIS lumen and wall ring overlays.

Technical Decisions Captured From The Prototype

- Use FC51 sharp-kernel CT for CPR/MPR detail. The earlier blur was mainly intrinsic to the FC03 standard/noise-reducing reconstruction, not a display bug.
- Use linear interpolation (order=1) for CT intensity reformats. At FC51 spacing this removes oblique staircase artifacts without materially blurring the already sharp reconstruc-

tion.

- Use nearest-neighbor display (`imshow(interpolation="nearest")`) for generated PNGs so no extra display-level smoothing is introduced.
- Use nearest-neighbor resampling only for labels, masks, or deliberately voxelated debug views.
- Keep separate lumen and vessel-wall point-cloud panels on the presentation slide when visibility matters; a combined point-cloud panel is useful as a compact alternative.
- Treat straightened-MPR contour overlays as projection envelopes unless exact plane/ring intersections are explicitly computed. The envelope is not the true outline of an oblique cut; it is the projected min/max silhouette of the ring set along the selected MPR chord.
- Keep the current projection-based rotation-minimizing frame for visualisation, but use a stricter double-reflection RMF in a future quantitative viewer².
- Freeze the `VesselScene` scene-bundle contract before building a larger app. The MEDIS pipeline, the voxel-mask pipeline, and the web viewer should meet at that contract rather than depending on each other's internal file formats.
- Use `.glb` for future browser-facing 3D surfaces and keep STL/GlTF only as intermediate or legacy research exports.

Coordinate Systems

Space	Units	Axes	Purpose	Transform rule
DICOM patient/world	mm	DICOM LPS unless converted	Physical reference for CT, MEDIS, masks, meshes	From DICOM tags or NIfTI affine/direction
Voxel index	index	(i, j, k)	Sampling source CTA/mask arrays	$\text{voxel} = D^{-1} \cdot (\text{world} - \text{origin}) / \text{spacing}$
CPR-cross	mm / station	(U, V, S)	Perpendicular cross-sections along vessel	$\text{world} = C[S] + U \cdot N[S] + V \cdot B[S]$
CPR-long	mm / station	(S, U, V)	Straightened MPR volume/viewer layout	Axis permutation of CPR-cross

Orientation policy:

- Preserve physical coordinates internally.
- DICOM CT is usually LPS; NIfTI written by `dcm2niix` may encode an axis flip in the direction/affine matrix.
- Screenshot display is a separate presentation choice. For the current slide, axial CTA is shown anterior-up, so the array view is vertically flipped together with the centerline over-

lay.

Mathematical Notation And Transforms

Use one explicit notation throughout the code and documentation:

- Patient/world point in millimetres: $\mathbf{p} = (x, y, z)^\top \in \mathbb{R}^3$.
- Voxel index, possibly fractional for interpolation: $\mathbf{i} = (i, j, k)^\top$.
- Image origin: $\mathbf{o}$.
- Voxel spacing: $\mathbf{s} = (s_i, s_j, s_k)^\top$.
- Direction matrix: $\mathbf{D} \in \mathbb{R}^{3 \times 3}$, orthonormal, usually from DICOM/NIfTI metadata.
- Centerline point at station s_k : $\mathbf{c}_k$.
- Local frame at station k : tangent $\mathbf{T}_k$, normal $\mathbf{N}_k$, binormal $\mathbf{B}_k$.

Voxel-to-world transform:

$$\mathbf{p} = \mathbf{o} + \mathbf{D} \text{diag}(\mathbf{s}) \mathbf{i}.$$

World-to-voxel transform:

$$\mathbf{i} = \text{diag}(\mathbf{s})^{-1} \mathbf{D}^\top (\mathbf{p} - \mathbf{o}).$$

This formula is mandatory. Ignoring $\mathbf{D}$ caused the original all-zero CPR sampling bug.

CPR-cross coordinate map:

$$\mathbf{p}(k, u, v) = \mathbf{c}_k + u \mathbf{N}_k + v \mathbf{B}_k.$$

Straightened MPR / rotational CPR map:

$$\mathbf{p}(k, r, \theta) = \mathbf{c}_k + r (\cos \theta \mathbf{N}_k + \sin \theta \mathbf{B}_k).$$

Intensity sampling:

$$I_{\text{cross}}(u, v, k) = I_{\text{CTA}}(\mathbf{p}(k, u, v)),$$

$$I_{\text{long}}(k, r, \theta) = I_{\text{CTA}}(\mathbf{p}(k, r, \theta)).$$

Implementation convention:

- CT intensities use linear interpolation and an out-of-bounds value of -1024 HU.
- Masks and labels use nearest-neighbor interpolation.
- PNG display uses nearest-neighbor display interpolation to avoid extra visual smoothing.
- NIfTI/DICOM orientation should follow the metadata internally; screenshot orientation is a labelled display choice^{3,4}.

Core Mathematical Algorithms

Centerline Parameterisation

Given raw centerline candidates $\mathbf{q}_0, \dots, \mathbf{q}_n$, first fit a smooth curve, then parameterise by arc length. For a discrete curve:

$$L_i = \sum_{j=0}^{i-1} \|\mathbf{q}_{j+1} - \mathbf{q}_j\|_2.$$

Sample a smoothed spline curve at uniform stations:

$$s_k = k \Delta s, \quad \mathbf{c}_k = \text{Spline}(s_k).$$

The spline should be fit before final arc-length resampling, otherwise annotation jitter or skeleton jaggedness is preserved in the frame⁵.

Tangents use central differences and normalization:

$$\mathbf{T}_k = \frac{\mathbf{c}_{k+1} - \mathbf{c}_{k-1}}{\|\mathbf{c}_{k+1} - \mathbf{c}_{k-1}\|_2}.$$

Endpoint tangents use one-sided differences.

Rotation-Minimizing Frame

Current prototype method: projection/Bishop step. Given $\mathbf{N}_k$ and the next tangent $\mathbf{T}_{k+1}$:

$$\tilde{\mathbf{N}} = \mathbf{N}_k - (\mathbf{N}_k \cdot \mathbf{T}_{k+1})\mathbf{T}_{k+1},$$

$$\mathbf{N}_{k+1} = \frac{\tilde{\mathbf{N}}}{\|\tilde{\mathbf{N}}\|_2}, \quad \mathbf{B}_{k+1} = \mathbf{T}_{k+1} \times \mathbf{N}_{k+1}.$$

This is acceptable for visualisation, but it can accumulate twist drift over long or highly curved vessels.

Future preferred method: double-reflection RMF². Define a reflection about vector $\mathbf{v}$:

$$R_{\mathbf{v}}(\mathbf{x}) = \mathbf{x} - 2 \frac{\mathbf{v} \cdot \mathbf{x}}{\mathbf{v} \cdot \mathbf{v}} \mathbf{v}.$$

Then propagate the frame:

$$\mathbf{v}_1 = \mathbf{c}_{k+1} - \mathbf{c}_k, \quad \mathbf{T}_k^L = R_{\mathbf{v}_1}(\mathbf{T}_k), \quad \mathbf{N}_k^L = R_{\mathbf{v}_1}(\mathbf{N}_k),$$

$$\mathbf{v}_2 = \mathbf{T}_{k+1} - \mathbf{T}_k^L, \quad \mathbf{N}_{k+1} = R_{\mathbf{v}_2}(\mathbf{N}_k^L), \quad \mathbf{B}_{k+1} = \mathbf{T}_{k+1} \times \mathbf{N}_{k+1}.$$

If $\|\mathbf{v}_1\|$ or $\|\mathbf{v}_2\|$ is numerically tiny, fall back to copying or re-orthogonalizing the previous frame. This method is the target for quantitative exports.

Centerline From Voxel Masks

For a filled lumen or filled vessel volume M , compute the Euclidean distance transform $d(\mathbf{x})$. A fast-marching medial path solves:

$$\|\nabla T(\mathbf{x})\| = P(\mathbf{x}), \quad P(\mathbf{x}) = \frac{1}{d(\mathbf{x}) + \varepsilon}.$$

Here P is the traversal cost/slowness, so high-radius medial voxels have low cost. The centerline is recovered by following the steepest descent path on T from distal seed to proximal seed, biased toward high-distance medial voxels^{6,7}.

Fallback: 3D thinning/skeletonization, graph extraction, endpoint-constrained longest or lowest-cost path, then spline smoothing⁸.

Surface Extraction From Masks

For binary masks, extract the isosurface at level 0.5 using marching cubes⁹. Vertices must then be mapped to world coordinates with the voxel-to-world transform above.

Mesh post-processing:

- split connected components and label them as lumen/wall by centerline containment and normal direction;
- Taubin smooth with non-shrinking parameters, e.g. $\lambda = 0.50$, $\mu = -0.53$, about 20 iterations¹;
- decimate with quadric error metrics toward a web budget, e.g. 8k-12k triangles per surface¹⁰;
- recompute normals and preserve scalar fields such as wall thickness or plaque class.

Core Viewer Requirements

1. Data loading

- Load CTA volume with correct affine/direction handling.
- Load MEDIS contours, mesh files, NIfTI masks, and metadata.
- Preserve patient/world coordinates instead of assuming identity direction.

2. Geometry model

- Smooth the raw centerline candidates, for example MEDIS lumen centroids, with a spline first.
- Reparameterize the smoothed curve by arc length and resample at the target station spacing.
- Compute a stable rotation-minimizing frame along the centerline.

- Future preferred method: double-reflection RMF for reproducibility and lower accumulated twist than simple projection propagation.
- Initial frame should eventually have clinical meaning: project an anatomical direction such as anterior or an RAO/LAO reference into the plane perpendicular to the first tangent. The current “axis least aligned with tangent” rule is acceptable for prototype visualisation but arbitrary.
- Store transforms between world, voxel, CPR, and straightened coordinates.

3. Views

- Axial CTA with correct radiological orientation.
- 3D view: point cloud, mesh, centerline, voxel-mask surface.
- CPR cross-section view perpendicular to centerline.
- Straightened MPR with rotatable angle around centerline.
- Straightened MPR contour mode:
 - envelope overlay for presentation,
 - exact intersection overlay for quantitative review.
- Linked cursor contract:
 - a single integer station index s is the shared state;
 - cross-section view displays CPR slice s ;
 - straightened MPR displays a vertical cursor at column s ;
 - 3D view highlights `centerline[s]`;
 - axial CTA jumps to the source slice closest to `centerline[s]`.

4. Visual overlays

- MEDIS lumen and vessel-wall rings.
- Mesh surfaces for lumen and vessel wall.
- Centerline and station markers.
- Voxel mask contours / marching-cubes surfaces.
- Plaque components if available: calcified, non-calcified, low attenuation.

MEDIS-Based Pipeline

1. Parse raw MEDIS contour .txt.
2. Extract lumen centroids as the initial centerline samples.
3. Smooth the centroid path with a spline to suppress ring-to-ring annotation jitter.
4. Resample the smoothed curve by arc length.
5. Compute rotation-minimizing frame.
6. Generate CPR volume by sampling CTA in world coordinates.
7. Project each MEDIS ring into local CPR coordinates for validation overlays.
8. Build meshes from contour rings for 3D rendering.

Voxel-Mask / Vendor Plaque-Mask Pipeline

Future input may be a voxel mask representing the vessel wall/plaque volume between inner and outer boundaries. “Vendor plaque mask” is used generically here; a specific AutoPlaque/Clearly-style export should be handled according to its actual label semantics.

The canonical input set should be:

- required: {CTA, plaque-annulus mask};
- optional: {lumen mask, outer-wall mask, centerline polyline, endpoint seeds}.

The pipeline must degrade gracefully across these input shapes. A mask-only annulus can provide outer wall, inner lumen boundary, and an estimated centerline, but failures should fall back to a supplied centerline or endpoint-guided routing.

Expected handling:

1. Load {CTA, mask} and verify physical-space compatibility.
 - Compare spacing, origin, direction/affine, and orientation within tolerance.
 - If the mask is aligned but sampled differently, resample the mask to CTA space with nearest-neighbor interpolation.
 - Stop if the metadata implies unresolved registration mismatch.
2. Detect mask topology.
 - Classify as annulus, solid blob, disconnected components, or multi-branch structure.
 - For disconnected components, use connected-component filtering or explicit vessel selection.
 - Reject or flag non-annular masks if lumen/wall separation is required and cannot be inferred.
3. Recover or import the centerline, in priority order.
 - **Default:** distance-transform / fast-marching medial path. Fill the annulus to obtain a full vessel volume, compute a Euclidean distance transform, then route a minimal-cost path with cost approximately $1 / (\text{distance} + \epsilon)$ so the path follows the medial high-radius ridge^{6,7,11}.
 - **Fallback:** voxel skeleton to graph to longest or endpoint-constrained path; then spline smooth and arc-length resample.
 - **Manual escape hatch:** accept a vendor-provided centerline, uploaded polyline, or two clicked endpoints plus auto-routing.
 - Validate that each centerline station remains safely inside the filled vessel volume; reject paths whose minimum distance-to-boundary is below a vessel-radius threshold, e.g. < 1 mm.
4. Smooth and resample the accepted centerline.
 - Fit a cubic spline or equivalent smooth curve before arc-length resampling.
 - Compute local tangent and RMF.
 - Store all world/voxel/CPR transforms in the scene bundle.
5. Extract surfaces from the annulus mask.
 - Run marching cubes at $\text{level}=0.5$ in voxel space, then transform vertices to world coordinates using $\text{world} = \text{origin} + \text{direction} \cdot (\text{spacing} \cdot \text{voxel})$.
 - Split the resulting mesh by connected components. A binary annulus should produce inner and outer surface components.

- Label components as lumen and wall using centerline containment/distance and normal direction checks.
 - Smooth with Taubin smoothing, not plain Laplacian smoothing, to reduce staircase artifacts without shrinking the vessel¹.
 - Decimate each surface to a target web budget, for example 8k-12k triangles per surface.
 - Compute normals and optional per-vertex colors, e.g. wall thickness or plaque class.
6. Generate CPR/MPR using the same centerline/frame machinery as the MEDIS pipeline.
 - CT intensities use linear interpolation.
 - Mask labels use nearest-neighbor interpolation.
 - CPR-space label maps can then produce cross-section contours and straightened-MPR overlays.
 7. Compute per-station quantitative metrics.
 8. Write the common `VesselScene` scene bundle.

This pipeline is intentionally parallel to the MEDIS path. MEDIS builds surfaces from contour rings; the mask path builds surfaces from marching cubes. Downstream views, exports, validation reports, and quantitative metrics should not care which path created the scene.

Optional Interactive Segmentation Track

The older `vessel-segmenter.md` plan describes a third source path: start from CTA only, let the user place two endpoint clicks, and estimate centerline, lumen, and outer wall with classical image processing. This is not required for the current presentation slide or for vendor-mask ingestion, but it is valuable as a fallback and as a future research feature.

Source input:

- CTA volume;
- two user-selected endpoints, e.g. proximal and distal LAD;
- optional vesselness map cache.

Pipeline:

1. Compute a vesselness image, e.g. Frangi/Hessian multi-scale filter.
2. Build a cost map where vessel-like voxels have low traversal cost.
3. Extract an initial centerline using FMM or minimal path between the endpoint seeds.
4. Generate perpendicular MPR cross-sections along that centerline.
5. Convert each cross-section into polar coordinates around the center point.
6. Use dynamic programming / graph search to find lumen and outer-wall radial boundaries.
7. Refine the centerline from lumen centroids to reduce corner cutting.
8. Smooth, arc-length resample, compute RMF, build surfaces, CPR volumes, overlays, and metrics.
9. Emit the same `VesselScene` bundle as the MEDIS and mask pipelines.

This track is more algorithmically ambitious than viewing a supplied mask. It should therefore be treated as optional and interactive, with visible confidence/validation outputs. It is useful

when no MEDIS contours and no vendor plaque mask are available.

Scene Bundle Format

A single vessel export should be a portable zip archive named like:

vessel_scene_{patient}_{vessel}.zip

Recommended layout:

```
manifest.json
cta.nii.gz          # optional full CT, or external reference
mask.nii.gz         # source annulus/label mask audit trail
centerline.json     # points, tangents, normals, binormals, arc lengths
cpr/
  cross_volume.nii.gz  # U x V x S CPR cross-section volume
  long_volume.nii.gz   # S x U x V straightened/rotational source
  frame.json          # redundant frame copy for standalone use
surfaces/
  lumen.glb          # inner/lumen surface
  wall.glb           # outer vessel-wall surface
  plaque_components.glb # optional plaque labels or components
overlays/
  medis_rings.json    # if MEDIS source
  voxel_mask_uv.json  # per-station projected mask contours
quant/
  per_station_metrics.json
  lesion_summary.json
validation.json
preview.png
```

Format choices:

- .glb for meshes because it is web-native, compact, single-request, and supports normals, materials, and per-vertex colors. Prefer it over STL for the future app.
- .nii.gz for CT/CPR volumes because it preserves affine metadata and is supported by NiiVue.
- JSON for centerline, frame, validation, and quantitative metrics because it is inspectable and browser-native.
- One zip for upload/download ergonomics and reproducible audit trails.

Quantitative Outputs

The first viewer can remain visual, but the future quantitative review mode should always export explicit per-station and lesion-level outputs. CPR image-space contours are useful for display, but the preferred quantitative route is plane/surface intersection: intersect the lumen surface with the perpendicular plane at station k , recover a ring in (u, v) coordinates, then compute the following metrics from that ring.

Lumen area:

$$A_k = \frac{1}{2} \left| \sum_{m=1}^M u_m v_{m+1} - u_{m+1} v_m \right|$$

with cyclic indexing $M + 1 \equiv 1$.

Effective lumen diameter:

$$D_{\text{eff},k} = 2 \sqrt{\frac{A_k}{\pi}}.$$

Minimum lumen diameter and minimum lumen area:

$$\text{MLD} = \min_{k \in \mathcal{L}} D_{\text{min},k}, \quad \text{MLA} = \min_{k \in \mathcal{L}} A_k,$$

where $\mathcal{L}$ is the analysed lesion or vessel segment. $D_{\text{min},k}$ should be documented as a Feret/minimum caliper diameter or as another explicit diameter definition.

Area stenosis:

$$\text{AS} = 1 - \frac{A_{\text{min}}}{A_{\text{ref}}}, \quad A_{\text{ref}} = \frac{A_{\text{ref,prox}} + A_{\text{ref,dist}}}{2}.$$

Diameter stenosis:

$$\text{DS} = 1 - \frac{D_{\text{min}}}{D_{\text{ref}}}.$$

Reference area/diameter selection must be recorded, because this choice changes the reported stenosis percentage^{12,13}.

Wall thickness for an inner-surface point $\mathbf{v}_{\text{in}}$:

$$t(\mathbf{v}_{\text{in}}) = \min_{\mathbf{v}_{\text{out}} \in S_{\text{outer}}} \|\mathbf{v}_{\text{out}} - \mathbf{v}_{\text{in}}\|_2.$$

Per-station wall thickness is the mean, median, maximum, and angular profile of all inner vertices in a tangent band around station k .

Plaque and calcium volume:

$$V_{\text{label}} = N_{\text{voxels,label}} s_i s_j s_k.$$

Calcium volume from CTA can be counted inside the plaque/wall mask using a threshold such as $\text{HU} > 130$. An Agatston-style score requires a non-contrast calcium series and slice-specific

rules:

$$\text{Score} = \sum_{\text{slices}} \sum_{\text{lesions}} A_{\text{lesion}} w(\max \text{HU}),$$

with weights $w = 1, 2, 3, 4$ for maxima in 130-199, 200-299, 300-399, and ≥ 400 HU, respectively¹⁴. CCTA plaque attenuation categories should be documented separately from Agatston scoring¹⁵.

Export:

- per-station area, diameter, wall thickness, plaque burden, and calcium flags;
- lesion-level MLD, MLA, stenosis percentages, lesion length, plaque volumes, and reference choices;
- CSV/JSON with units and method metadata.

Mathematical Validation Metrics

Direction-matrix check: all world-to-voxel sampling must use

$$\mathbf{i} = \text{diag}(\mathbf{s})^{-1} \mathbf{D}^\top (\mathbf{p} - \mathbf{o}).$$

Frame unit-length error:

$$\varepsilon_{\text{unit}} = \max_k \max (|\|\mathbf{T}_k\|_2 - 1|, |\|\mathbf{N}_k\|_2 - 1|, |\|\mathbf{B}_k\|_2 - 1|) < 10^{-6}.$$

Frame orthogonality error:

$$\varepsilon_{\perp} = \max_k \max (|\mathbf{T}_k \cdot \mathbf{N}_k|, |\mathbf{T}_k \cdot \mathbf{B}_k|, |\mathbf{N}_k \cdot \mathbf{B}_k|) < 10^{-6}.$$

Right-handedness:

$$\min_k \det[\mathbf{T}_k, \mathbf{N}_k, \mathbf{B}_k] > 0.999.$$

Centerline CT coverage:

$$\frac{\#\{k : \mathbf{c}_k \in \Omega_{\text{CTA}}\}}{N} = 1.0.$$

MEDIS ring out-of-plane residual:

$$\rho_i = \max_{\mathbf{p} \in R_i} |(\mathbf{p} - \mathbf{c}_{k(i)}) \cdot \mathbf{T}_{k(i)}| < 1 \text{ mm}.$$

Mask/CTA affine agreement:

$$\varepsilon_{\text{aff}} = \|\mathbf{o}_M - \mathbf{o}_{CT}\|_2 + \|\mathbf{D}_M - \mathbf{D}_{CT}\|_F + \|\mathbf{s}_M - \mathbf{s}_{CT}\|_2.$$

If this exceeds the chosen tolerance, resample only when a valid transform is known; otherwise stop.

Centerline-in-mask check:

$$\min_k d(\mathbf{c}_k, \partial M) > r_{\min}, \quad r_{\min} = 1 \text{ mm}$$

for mask-derived centerlines.

Surface checks:

- non-manifold edge count equals 0;
- connected-component count matches expected topology;
- lumen/wall surfaces are watertight if surface-based quantification is enabled;
- vertex and triangle counts are reported before and after smoothing/decimation.

Twist-drift diagnostic:

$$\Delta\phi = \max_k \angle (\mathbf{N}_k^{\text{projection}}, \mathbf{N}_k^{\text{double-reflection}}).$$

If $\Delta\phi > 5^\circ$ on a long vessel, e.g. RCA, use double-reflection RMF for exported quantitative views.

Straightened MPR contour overlays must be documented as projection envelopes unless exact plane/ring or plane/surface intersections are computed.

Synthetic phantoms should include non-identity direction matrices, anisotropic spacing, known lumen/wall radii, and at least one curved tube where the expected CPR coordinates are known.

Failure Modes And Fallbacks

Failure mode	Detection	Fallback
Centerline exits CT volume	any station outside source image bounds	trim stations, warn, or require source series correction
Missing FC51/sharp reconstruction	no matching series/kernel found	use FC03 but label as standard-kernel context, not sharp plaque-detail view
Mask lacks lumen/wall labels	label schema missing or single foreground only	show mask surface, defer lumen/wall measurements, request label semantics
Skeleton branch ambiguity	multiple graph paths with similar cost	require seed points or manual vessel selection

Failure mode	Detection	Fallback
Frame twist or instability	high station-to-station frame rotation	recompute with double-reflection RMF and smoothing
MEDIS and CT misregistration	ring residuals or bbox check fail	stop quantitative export and show registration diagnostic
Mask and CTA mismatch	spacing/origin/direction differ beyond tolerance	resample only if affine alignment is explicit; otherwise stop
Non-annular mask	topology has no lumen hole or ambiguous components	require lumen label, centerline, or manual confirmation
Bifurcation inside mask	graph has branch points near target vessel	one-vessel-at-a-time path selection or endpoint seeds
Total/subtotal occlusion	lumen hole disappears or distance field collapses	accept user centerline, mark local measurements unreliable
Stent or blooming artifact	very high HU and irregular lumen boundary	preserve visualisation but flag quantification caveat
Centerline leaves lumen/vessel volume	distance-to-boundary falls below threshold	reject centerline and fall back to alternate path/manual seeds
Surface not watertight	mesh boundary edges or invalid normals	repair mesh if possible; otherwise disable surface-based quant

Performance Targets

Initial targets for an interactive viewer:

- CPR/MPR generation for one vessel: < 2 s after source volume is loaded.
- Full interactive CTA-only segmentation from two endpoint clicks: initial target 20-60 s per vessel on CPU.
- Linked station cursor update: < 50 ms.
- 3D mesh or mask first paint: < 100 ms for precomputed surfaces.
- Scene memory for one vessel and one CTA series: < 500 MB.
- Exported presentation slide generation: < 10 s.

These budgets influence whether the implementation should use NiiVue, VTK.js, Three.js, PyVista export, or precomputed PNG/NIfTI assets.

Web Architecture Options

Architecture	Compute location	Pros	Cons
Backend service	Python server, e.g. FastAPI worker	deterministic, fast, easy to cache, easy to log validation reports	requires hosting and upload of large CTA/mask files
Pure client	TypeScript plus ITK- WASM/Pyodide/Web Worker	static deployment, no server-side patient data handling	restrictive in practice for this workload: slow large-volume compute, browser memory ceiling, difficult debugging/validation
Hybrid scene-bundle workflow	backend/preprocessor emits bundle; browser renders	best practical UX: heavy compute once, fast interactive viewer	two pieces to maintain

Recommended production path:

1. Python backend or local preprocessing CLI accepts `{cta.nii.gz, mask.nii.gz}` or MEDIS contours.
2. It emits the frozen `VesselScene` bundle.
3. Browser viewer loads the bundle.
4. NiiVue renders axial CTA, CPR cross-sections, and straightened MPR NIfTI volumes.
5. Three.js renders `.gltf` lumen/wall/plaque surfaces.
6. A single shared `station_index` drives all panels.

For this project, do not put heavy preprocessing in the browser. Earlier browser-compute experiments with ITK-WASM/Pyodide/Web Workers were too restrictive for large CTA/mask workflows. Keep the browser as renderer, linked-view controller, and lightweight interaction layer. Use Python, either as a local CLI or backend service, for scene-bundle generation.

Plain TypeScript feasibility:

- The viewer itself can absolutely be plain TypeScript: NiiVue for NIfTI/CPR volumes, Three.js for `.gltf` surfaces, a small state store for `station_index`, and JSON loaders for centerline/quant/validation.
- Light geometry can also be TypeScript: drawing overlays, interpolating along a precomputed centerline, changing MPR angle if the CPR-long volume is already present, and loading/slicing precomputed label maps.
- Heavy preprocessing should stay out of the browser. Marching cubes, distance transforms, FMM, Frangi vesselness, and large-volume resampling are technically possible with ITK-WASM/Pyodide/Web Workers, but previous tests were restrictive enough that they should not be the planned path.
- Production direction: Python local CLI/backend creates the scene bundle; the app frontend remains TypeScript-only.

Validation Report Targets

For every vessel/viewer export, produce a small machine-readable validation summary:

- CT series UID, reconstruction kernel, spacing, origin, direction.
- Vessel name, source type, contour/mask counts.
- Mask topology class: annulus, solid, disconnected, branched, or rejected.
- Mask/CTA alignment error or resampling transform.
- Centerline length and number of stations.
- Frame unit-length, orthogonality, and handedness errors.
- Fraction of centerline stations inside the source CT volume.
- Fraction of centerline stations inside the mask or filled vessel volume.
- Minimum and percentile distance-to-mask-boundary along the centerline.
- Surface watertightness, component count, and vertex/triangle counts before and after decimation.
- MEDIS ring-to-station distance statistics.
- Interpolation mode for CT intensities and masks.
- Axial display convention used in screenshots.
- Git commit or archive hash of the code.
- Algorithm versions: centerline method, smoothing parameters, RMF method, CPR spacing, interpolation order.
- Window/level, display interpolation, and screenshot timestamp.

Future Implementation Plan

1. Refactor current scripts into reusable modules:
 - `io.py`
 - `centerline.py`
 - `frame.py`
 - `cpr.py`
 - `surfaces.py`
 - `viewer.py`
2. Add a structured vessel model:
 - implement the `VesselScene` schema;
 - patient id and vessel name;
 - source type: MEDIS, mask, or hybrid;
 - centerline and RMF frame;
 - lumen, wall, and optional plaque surfaces;
 - labels and CPR-space overlays;
 - quantitative metrics;
 - provenance metadata.
3. Add tests with synthetic phantoms:
 - straight cylinder,
 - curved tube,
 - oblique volume orientation,
 - known lumen/wall radii.

- branched mask with a distractor branch,
 - anisotropic voxel spacing.
4. Build scene-bundle export:
 - manifest.json;
 - NIfTI CPR volumes;
 - .gltf surfaces;
 - overlay JSON;
 - quantitative JSON/CSV;
 - validation report;
 - preview PNG.
 5. Build an interactive viewer:
 - NiiVue for NIfTI volumes and slices,
 - Three.js for .gltf 3D surfaces,
 - linked station slider,
 - angle slider for straightened MPR.
 6. Add export modes:
 - single presentation slide,
 - appendix figures,
 - interactive HTML,
 - small validation report.
 7. Optional later: CTA-only interactive segmentation:
 - endpoint-click UI;
 - vesselness/FMM centerline extraction;
 - polar dynamic-programming lumen/wall segmentation;
 - centerline refinement from lumen centroids;
 - export as the same VesselScene contract.

Open Decisions

- Whether clinical display should default to linear CT reformat or nearest-neighbor debug view.
- Whether straightened MPR contour lines should be exact plane intersections or projection envelopes.
- How vendor plaque-mask labels encode lumen, wall, and plaque classes.
- Whether the viewer should support multiple vessels simultaneously or one vessel per scene.
- Whether 4-D / gated CT support is in scope.
- Whether hover readout should include HU, physical coordinates, station index, and plaque label.
- Whether the default axial display convention should be radiological or neurological, and how this is labelled in exports.
- Whether bifurcations are handled natively or represented as one selected branch per scene.
- Whether users can edit the centerline with 3D handles or only provide endpoint seeds.
- Whether scene bundles embed the full CTA or reference it externally to reduce file size.

- Whether production compute is a local Python CLI, FastAPI plus worker/cache, or serverless batch. Browser-only ITK-WASM/Pyodide is not the preferred path for heavy preprocessing.
 - Whether .glb surfaces should store wall-thickness/plaque-class colors directly or reference separate scalar arrays.
 - Whether CTA-only endpoint-based segmentation belongs in the first app version or remains a research/fallback module.
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